LAB ASSIGNMENT

NLP Foundations: Handwriting OCR Evaluation

Deepening your understanding after Lab 6

Course	Artificial Intelligence
Practical	Lab 6 of 8 - Handwriting OCR Evaluation
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Purpose

This assignment is to be completed after the Lab 6 practical. You do not need to repeat the entire experiment. Instead, use what you learned to explain the concepts, interpret the results, reason about OCR errors and suggest improvements. Answer in your own words.

Instructions

1. Complete the assignment individually.
2. Use your Lab 6 practical outputs and the results discussed during the lab as evidence.
3. Show all calculation steps. Round numerical answers to 2–3 decimal places where appropriate.
4. You may refer to your lab results; you do not need to rerun the complete OCR experiment.
5. Include screenshots or small tables only where requested. Ensure that all evidence is readable.

Part A - Understand the concepts

Q1. NLP pipeline

Describe the five stages of an NLP pipeline: acquire, normalise, represent, model and evaluate. For each stage, give an example of what happens when a handwritten medicine-name image is processed.

Acquire	Collect the raw data
Normalise	Clean the data so it is consistent
Represent	Turn data into a form the model can use
Model	Apply the algorithm
Evaluate	Measure how good the output is

Q2. How OCR works

Explain the difference between segmentation and character recognition. Give one example of a failure at each stage. Why is it useful to identify which stage caused an error?

Segmentation finds where the text is and splits it into lines, words or characters. Failure example: two joined handwritten letters "rn" are treated as one shape, or the word is not detected at all, which gives an empty output.

Character recognition decides which character each segmented shape is. Failure example: the shape is found correctly, but an "m" is read as "d".

Each stage needs a different fix. Segmentation errors need better preprocessing while recognition errors need a better model or more training data.

Q3. Text normalisation

The reference is "Tab Azimax 40mg" and the OCR output is "tab azimax 40 mg". Apply the Lab 6 normalisation rule (lowercase and keep only letters and digits). Show both normalised strings. Why must the same rule be applied to both?

The rule is to lowercase, then keep only letters and digits.

Reference: "Tab Azimax 40mg" - tabazimax40mg

OCR output: "tab azimax 40 mg" - tabazimax40mg

If only one side is normalised, differences in formatting would be counted as OCR errors even though the text is the same. Using the same rule on both makes the comparison fair, so only real reading mistakes are measured.

Q4. Normalisation trade-off

The Lab 6 normaliser removes spaces. Describe one situation where removing spaces could hide an important difference. Suggest how you would change the normalisation rule for that situation.

Situation: dosage numbers. A reference of "2 50mg" (2 tablets of 50 mg) and an OCR output of "250mg" both normalise to 250mg. They look identical after normalisation, but in practice it is a dangerous dosage difference.

The same problem happens with punctuation: "2.5mg" and "25mg" both become 25mg.

Part B — Work with OCR measurements

Q5. Levenshtein distance

Calculate the Levenshtein distance between “breeky” (reference) and “beeky” (prediction). Identify the edit operation(s). Then calculate the normalised edit distance by dividing the distance by the length of the longer string.

Reference: b r e e k y (6 chars)

Prediction: b - e e k y (5 chars)

- Edit operation: **1 deletion**. The “r” was deleted.
- Levenshtein distance = **1**
- Normalised edit distance = 1 ÷ max (6, 5) = 1 ÷ 6 = **0.167**

Q6. Character Error Rate (CER)

An OCR system produces 4 substitutions, 2 deletions and 1 insertion. The reference contains 35 characters. Calculate CER using CER = (S + D + I) / N. Explain what the result means. Can CER be greater than 1? Explain briefly.

CER = (S + D + I) / N = (4 + 2 + 1) / 35 = 7 / 35 = 0.200

There are about 20-character errors for every 100 reference characters.

Can CER be greater than 1? Yes. Insertions are counted but N only counts reference characters.

Q7. Compare the metrics

Explain exact-match accuracy, mean normalised edit distance, CER and blank predictions. For each metric, state whether a higher or lower value is generally better. Why is exact-match accuracy alone insufficient?

Metric	Meaning	Better when
Exact-match accuracy	Fraction of images where the normalised output equals the reference exactly	Higher
Mean normalised edit distance	Average of (edit distance ÷ longer length) across all images	Lower
CER	Total character errors ÷ total reference characters	Lower
Blank predictions	Number of images where OCR returned nothing	Lower

Why exact match alone is not enough: An output of “azimx” (1 character wrong) and “zzz” (completely wrong) both score 0. It cannot show partial improvement, so a system can get much better while exact match stays the same.

Part C — Interpret the Lab 6 experiment

Use the results from Lab 6. Both pipelines were evaluated on the same 89 labelled images, with the same normalisation rule and fixed Tesseract configuration.

Q8. Read the results

Describe at least two improvements shown by the preprocessed results. Use values from the table as evidence.

1. CER decreased from 0.883 to 0.806. Fewer characters were wrong overall.
2. Blank predictions decreased from 42 to 15. Preprocessing helped Tesseract find text in 27 more images.
3. Mean normalised edit distance decreased from 0.869 to 0.785. On average, outputs were closer to the correct name.
4. Exact match increased from 0.000 to 0.011. 0.011 × 89 ≈ 1 image (“ostium”) fully correct.

Q9. Explain the result pattern

CER and mean normalised edit distance improved, and blank predictions decreased. However, exact-match accuracy increased only from 0.000 to 0.011. Explain how preprocessing can help recover text without making the complete medicine name correct.

-One wrong letter makes the whole name wrong. CER and edit distance give **partial credit** for every letter that gets closer. Preprocessing made the text **easier to find**, so Tesseract returned fewer blanks and more correct letters.

Q10. Controlled experiment

Why did the lab keep the Tesseract settings (--oem 1 --psm 7), the 89 images and the normalisation rule unchanged? What would be difficult to conclude if several settings changed at once?

The only thing that changed is **the image preprocessing**. The images, the Tesseract settings and the normalisation stayed the same, so any improvement **must come from the preprocessing**.

If several things changed at once, **we couldn’t tell which change caused the result**. Changes could also cancel each other out, or the score could look better just because the scoring changed.

Q11. Diagnose the failure

Identify the likely category for each case and explain:

(a) Tesseract raises an error while processing an image;

(b) OCR returns an empty string;

(c) OCR returns “nidu” when the reference is “nims”.

Case	Category	Explantion
(a) Tesseract raises an error	**Engine error**	The software failed on that image. It is recorded as a failure, not deleted.
(b) Empty string	**No text detected**	Tesseract couldn’t find the text.
(c) “nidu” instead of “nims”	**Character confusion**	It found the text but misread letters

Part D — Apply your understanding

Q12. Suggest improvements

Suggest TWO improvements you would investigate next to improve handwritten OCR. For each, explain which error it targets and how you would test whether it helps. Do not claim an improvement works without evidence.

1. Contrast enhancement and binarization which turns the image into clear black and white

- **Targets:** no text detected
- **Test:** add only this step, keep the same 89 images, settings and normalisation, then check whether blanks and CER go down.

2. A handwriting-trained OCR model

- **Targets:** character confusion
- **Test:** run it on the same 89 images with the same normalisation and compare exact match and CER.

Q13. Medical safety scenario

A clinic wants to use OCR to read medicine names and dosages from handwritten prescriptions. Explain at least THREE risks of using OCR output without human checking. Recommend a safer workflow.

1. **Wrong medicine**: one wrong letter can turn into a different drug.
2. **Wrong dosage**: "40mg" misread as "4mg" or "400mg" could harm the patient.
3. **Missing items**: 15 images still returned nothing, so a medicine could be skipped.

Safer workflow: OCR only makes a draft. The system flags uncertain results. A pharmacist checks every item against the original image and asks the doctor if anything is unclear. Every change is logged.

Q14. Reflection

What is the most important thing you learned from Lab 6? Identify one concept you initially found difficult and explain it using an example from the practical.

The most important thing I learned is that **finding text and reading text are different problems**. The concept I found difficult was why CER improved but exact match didn't. The prediction improved from blank to "Bveaky" with fewer errors and lower CER, but one wrong letter still makes exact match zero.

Submission to your AI Lab Git repository

IMPORTANT: Submit your completed assignment to the Git repository you created for the AI Lab.

Do not keep the assignment only on your computer or send it only through a private message. Upload it to your existing AI Lab Git repository.

6. Create a folder named Lab06_Assignment.
7. Save your completed report as Lab06_OCR_Assignment_YourName.pdf (or .docx).
8. Put supporting screenshots or figures in a subfolder named evidence.
9. Commit and push your changes to your existing AI Lab repository(if not done yet).
10. Open the repository online and confirm that your files are visible.

Suggested repository structure

```
AI-Lab-Repository/
└── Lab06_Assignment/
    ├── Lab06_OCR_Assignment_YourName.pdf
    └── evidence/
        ├── example_1.png
        └── results_table.png
```

Assessment focus

Your work will be assessed on conceptual understanding, correct calculations, interpretation of the actual Lab 6 results, quality of reasoning, clarity of writing and completeness of the Git repository submission.

Final reminder

The purpose is understanding, not simply reproducing code or copying results. Explain what the results mean, why the experiment was designed this way and what you would investigate next.